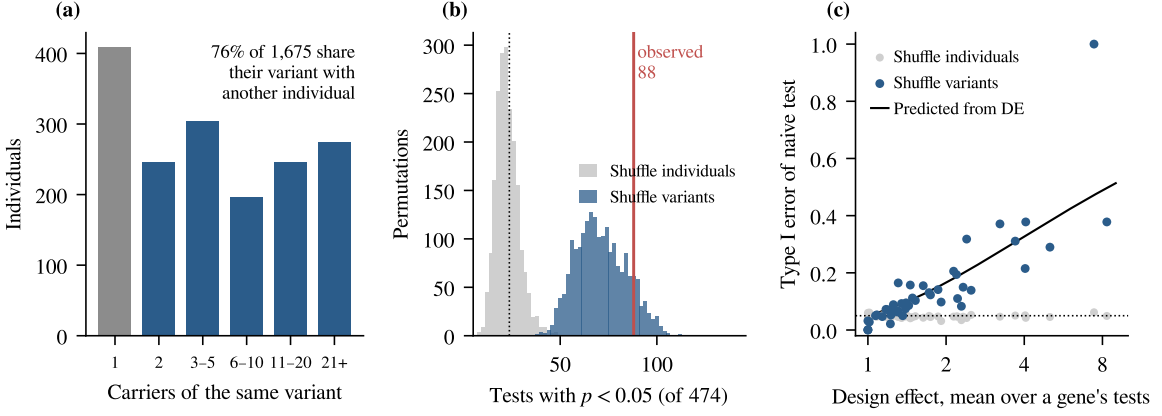
(a)
Individuals
400
300
200
100
0
76% of 1,675 share
their variant with
another individual
1
2
3–5
6–10
11–20
21+
Carriers of the same variant
(b)
Permutations
300
250
200
150
100
50
0
observed
88
Shuffle individuals
Shuffle variants
50
100
Tests with $p < 0.05$ (of 474)
(c)
Type I error of naive test
1.0
0.8
0.6
0.4
0.2
0.0
Shuffle individuals
Shuffle variants
Predicted from DE
1
2
4
8
Design effect, mean over a gene's tests